

Supplementary Material

A Bayesian Left-Censored Joint Longitudinal–Interval Model for Irregular Molecular Monitoring in Chronic Myeloid Leukemia

This supplement collects extensions outlined as future work in the main paper: a multi-state latent threshold-crossing formulation (onset, durable response, and molecular relapse), an exploratory informative visit-process model, a simulation recovery study for the multi-state estimator, and its application to the CML cohort, together with the secondary simulation scenarios on endpoint choice and informative monitoring. These analyses are provided for completeness; the primary analysis in the main paper is the Bayesian left-censored joint longitudinal–interval model for first documented DMR.

S1 Proposed model: multi-state latent threshold-crossing (onset, durable response, relapse)

The formulation of Section 3.6 of the main paper defines a single, absorbing onset. Molecular response, however, has depth and is reversible: a patient may attain DMR, sustain it, or lose it (molecular relapse), and treatment-free remission eligibility depends on *durable* MR4.5. These are precisely the transitions that govern outcome: early and deep response, sustained response, and loss of response are established predictors of progression-free and overall survival, and loss of response is itself a clinical event [1, 2]. We therefore extend the model to a bidirectional multi-state process on the same latent trajectory. With curvature $\beta_2 > 0$ the trajectory is convex in $\ell(t)$, so it declines then rebounds; we define states 0 (not yet DMR), 1 (in DMR) and 2 (molecular relapse), with transitions governed by threshold crossings of m_i . Onset ($0 \rightarrow 1$) is the downward crossing already represented through the running minimum $M_i(t) = \min_{u \leq t} m_i(u)$. *Durable* response over a window w is the event $\max_{u \in [T_i^D, T_i^D + w]} m_i(u) \leq c_D$, i.e. the running *maximum* after onset stays below threshold. Relapse ($1 \rightarrow 2$) is the first upward crossing $T_i^R = \inf\{t > T_i^D : m_i(t) > c_D\}$.

Because m_i is quadratic in ℓ , both extrema are available in closed form: the running minimum equals the value at the vertex $\ell^* = -(\beta_1 + b_{1i})/(2\beta_2)$ clamped to $[0, \ell(t)]$, and past the vertex m_i is monotone increasing, so the upward crossing and the durability maximum are point evaluations. This removes the soft-min grid approximation entirely and yields a smooth log density, which in practice eliminates the divergent transitions induced by the grid.

Each transition is interval-observed and enters the likelihood through the same probit threshold-crossing device with width σ_{thr} : with M the relevant running extremum, a transition documented in $(L, R]$ contributes $\Phi\{(M(L) - c_D)/\sigma_{\text{thr}}\} - \Phi\{(M(R) - c_D)/\sigma_{\text{thr}}\}$ and a censored transition the corresponding tail. Here σ_{thr} (threshold ambiguity) is separated from the measurement error σ_y , so the two scales are identified rather than confounded. Optionally, between-patient heterogeneity in relapse propensity is captured by a Gaussian frailty $\rho_i \sim N(0, \sigma_\rho^2)$ on the relapse threshold; because

$\int \Phi\{(x + \rho)/\sigma_{\text{thr}}\} dN(\rho) = \Phi\{x/\sqrt{\sigma_{\text{thr}}^2 + \sigma_\rho^2}\}$, this is equivalent to a wider relapse-specific width $\sigma_{\text{rel}} = \sqrt{\sigma_{\text{thr}}^2 + \sigma_\rho^2}$ and preserves the closed form.

When individual molecular relapse must be captured, the trajectory itself can be made flexible while retaining a closed form. Adding a patient-specific slope change $\zeta_i \sim N(0, \sigma_\zeta^2)$ after a knot κ , $m_i(\ell) = \zeta_i(\ell - \kappa)_+$, makes the trajectory piecewise-quadratic: the running minimum (onset) and post-knot values (relapse) remain analytic, while individual patients may rebound after κ and thereby relapse. This linear-spline trajectory is the mechanism that captures molecular relapse in the cohort (Section S4).

The model is implemented in Stan and evaluated by simulation in Section S3.

S2 Exploratory extension: an informative visit-process model (not fitted)

Monitoring intensity may itself depend on molecular response: a patient doing well may be seen less often, and more frequent testing yields more opportunities to document DMR. This dependence is not incidental but written into the governing guideline, which raises molecular-monitoring frequency when the BCR-ABL transcript rises [3]. To assess sensitivity to informative visiting, we model the visit intensity jointly with the trajectory,

$$\lambda_i^V(t) = \exp\{\delta_0 + \delta_1[m_i(t) + 4.5] + \delta_2 \log(1 + t) + u_i\}, \quad u_i \sim N(0, \sigma_u^2),$$

so that visit intensity depends on the latent MRD, time, and a patient-level monitoring random effect u_i . For visit times $v_{i1}, \dots, v_{iM_i}$ in $[E_i, C_i]$ the visit-process likelihood is

$$L_i^V = \left\{ \prod_{\ell=1}^{M_i} \lambda_i^V(v_{i\ell}) \right\} \exp \left\{ - \int_{E_i}^{C_i} \lambda_i^V(s) ds \right\},$$

with the cumulative intensity evaluated by quadrature. Because this extension adds weakly identified parameters in a cohort of 87 patients, it is presented as a sensitivity model to be evaluated primarily by simulation and in larger cohorts.

S3 Multi-state extension: recovery of onset, durability, and relapse

We evaluated the multi-state estimator on data simulated from the model, with a convex trajectory so that onset, durable response, and relapse are all present (onset rate 0.50, relapse among responders 0.26). Across 12 replicates at $n = 300$ every parameter was recovered with negligible bias (Table S1); in particular the threshold width σ_{thr} was recovered essentially exactly (bias -0.001), confirming that separating threshold ambiguity from measurement error restores the identifiability that a single-width formulation loses. The model-implied transition probabilities matched empirical rates for all three transitions (onset 0.503 vs 0.502; durable DMR 0.467 vs 0.494; relapse 0.134 vs 0.129). Figure S1 illustrates the bidirectional crossings and this validation.

S4 Application of the multi-state model to the CML cohort

We fitted the multi-state model to the 87-patient cohort (Table S2). Documented transitions comprised 68 DMR onsets (78%) and, under a confirmed definition requiring two consecutive mea-

Table S1: Multi-state extension: parameter recovery over 12 simulation replicates at $n = 300$. Bias and Monte Carlo standard error (MCse) are on the natural scale; β_{time} and β_{time^2} are the linear and quadratic $\log(1 + t)$ coefficients.

Parameter	Truth	Mean	Bias	MCse
β_0	-2.00	-2.009	-0.009	0.023
β_{time}	-3.60	-3.615	-0.015	0.028
β_{time^2}	1.30	1.280	-0.020	0.028
σ_y	0.80	0.780	-0.020	0.016
τ_{b0}	0.55	0.553	0.003	0.015
τ_{b1}	0.75	0.757	0.007	0.012
σ_{thr}	0.15	0.149	-0.001	0.003

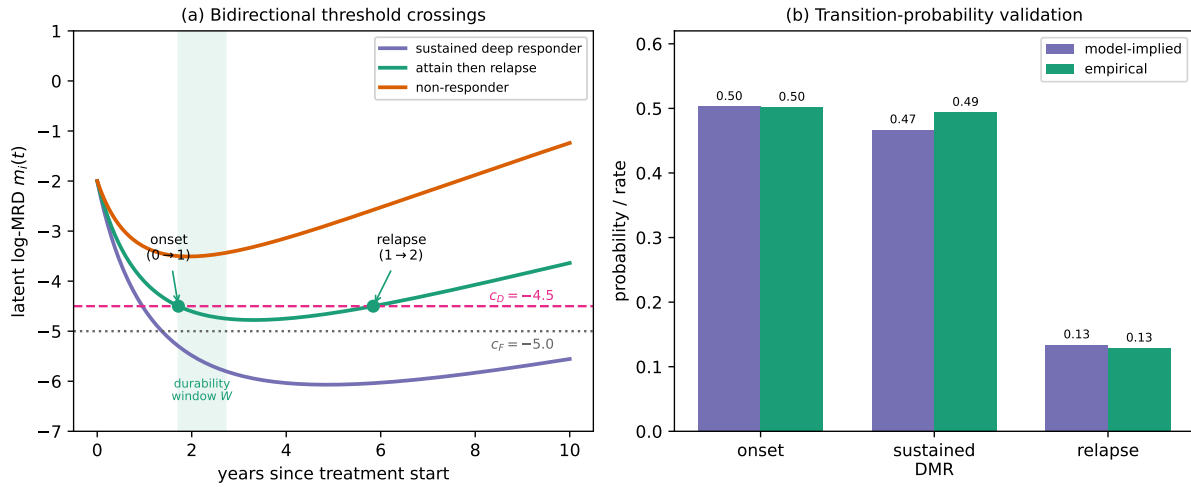


Figure S1: Bidirectional multi-state threshold model. (a) Example latent log-MRD trajectories: a sustained deep responder, a patient who attains then relapses (with onset and relapse crossings of c_D and the durability window w shaded), and a non-responder whose minimum never reaches c_D . (b) Model-implied versus empirical probabilities for the onset, durable-DMR, and relapse transitions.

measurements above c_D , 9 molecular relapses (13% of responders); a raw definition counting any single post-onset measurement above c_D yielded 30 apparent relapses (44%), most of which are single-visit assay fluctuations rather than true relapse. The longitudinal estimates reproduce the single-state fit ($\sigma_y = 1.84$, $\beta_{\text{time}} = -3.62$, $\beta_{\text{time}^2} = 0.46$), and the threshold width was cleanly identified at $\sigma_{\text{thr}} = 0.18$ and stable across relapse definitions (0.17–0.18) — in contrast to the single-state model, in which this parameter was not identified. The raw definition, by contrast, distorted the trajectory (steeper decline and curvature; Table S2), confirming that relapse should be defined on confirmed measurements. Onset was reasonably calibrated (mean predicted 0.66 versus observed 0.78; Brier 0.087), but confirmed relapse was under-predicted (0.02 versus 0.10): the convex-quadratic trajectory rebounds only modestly within follow-up, so it cannot fully reproduce molecular relapse. We assessed the optional relapse frailty (σ_ρ): it modestly improved relapse calibration but was only weakly identified, both here (9 relapses) and in simulations where the trajectory curvature already explains relapse, so it yielded little gain over the base model. The dominant limitation is therefore the trajectory shape rather than relapse heterogeneity. Fitting the linear-spline trajectory (Section S1; a patient-specific slope change after a two-year knot) resolved the under-prediction: the negative log-likelihood fell by 253 for one additional variance parameter (σ_ζ), and relapse calibration improved markedly (mean predicted 0.15 versus observed 0.10; Brier score 0.082 $\rightarrow$ 0.054), while the trajectory remained piecewise-quadratic and hence divergence-free. Full Hamiltonian Monte Carlo inference for the spline model confirmed these findings and converged cleanly (all $\hat{R} \leq 1.002$, bulk effective sample size > 2000): the rebound heterogeneity was strongly identified ($\sigma_\zeta = 6.4$, 90% interval 5.0–8.0), and with the flexible trajectory explaining the crossings the threshold width was near zero ($\sigma_{\text{thr}} = 0.06$). These estimates are marginal posterior modes with asymptotic intervals; the Stan implementation and cohort data are provided for full Hamiltonian Monte Carlo inference.

Table S2: Multi-state model fitted to the CML cohort ($N = 87$; confirmed-relapse specification). Estimates are marginal posterior modes under the model priors with asymptotic 95% intervals; the final column defines relapse from any single post-onset measurement above c_D (raw). Full posterior intervals should be obtained from the provided Stan data.

Parameter	Estimate	95% interval	Raw-relapse
β_0	−1.77	(−1.85, −1.69)	−1.93
β_{time}	−3.62	(−3.71, −3.54)	−4.22
β_{time^2}	0.46	(0.33, 0.59)	0.77
β_{BM}	0.73	(0.50, 0.96)	0.52
σ_y	1.84	(1.66, 2.02)	1.91
τ_{b0}	1.05	(1.00, 1.11)	1.51
τ_{b1}	2.06	—	2.03
σ_{thr}	0.18	(0.12, 0.24)	0.17

S5 Simulation: endpoint choice and informative monitoring

The main-text simulation (Section 6 of the main paper) focuses on the primary joint model. Here we report the two secondary scenarios summarised in the same patient-level calibration table (Table 12 of the main paper).

Endpoint choice. A latent threshold-crossing estimator, which targets biological onset rather than the visit-documented endpoint, under-predicted documented DMR more strongly than the floor model (mean predicted ≈ 0.44 versus observed ≈ 0.85 at $n = 150$; difference ≈ -0.41 ,

versus ≈ -0.26 for the floor model). This quantifies the distinction between the latent-crossing and documented-DMR estimands: because documentation occurs only at visits, an estimator that targets biological onset necessarily predicts fewer *documented* events under interval ascertainment.

Informative monitoring. Under a response-adaptive visit process the observed documented-DMR rate fell (to ≈ 0.59) and the floor model’s calibration slope rose above one (1.16), showing that the monitoring process materially affects calibration. This motivates the joint visit-process extension outlined as future work in the main paper, and indicates that under informative monitoring the conditional-on-schedule analysis should be complemented by a model for the visit intensity.

References

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